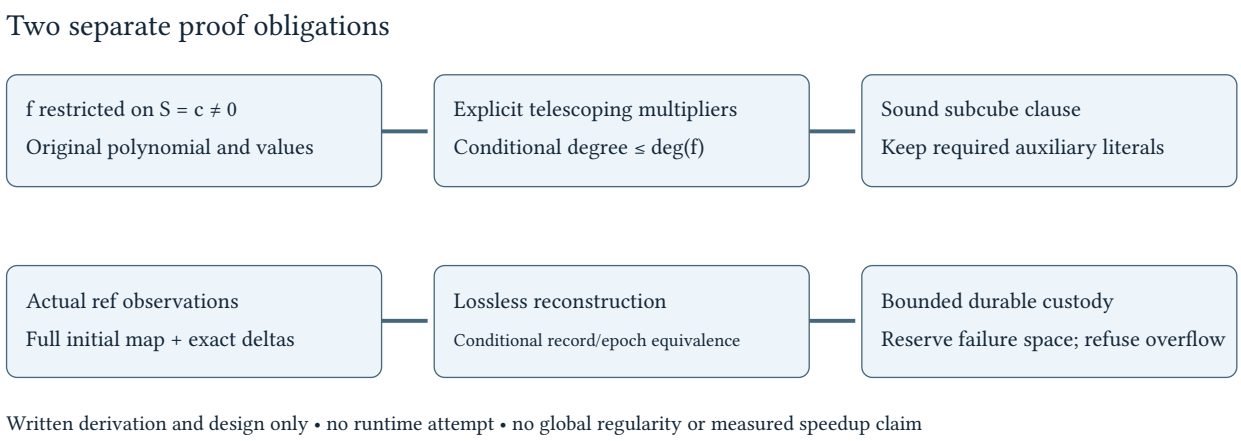
Arbitrary-subcube certificates and compact validation custody

Status: written derivation and design with independent static critique and corrections disposed. No candidate, oracle, metadata phase, replay or empirical comparison was executed.



The mathematical construction extends the fixed-prefix scalar certificate to any restriction set S. If a generator f becomes c≠0, successively substituting variables in S gives explicit multipliers h_r with f-c = sum_r (x_(i_r)-a_(i_r))h_r. For D=deg(f), each nonzero multiplier has degree at mostD-1, so the conditional refutation has degree at mostD. The fixed-value equations are essential to that conclusion.

The Boolean subcube indicator chi instead gives chi*f=c*chi modulo Boolean field equations. Its displayed product degree is bounded by D+|S|. The corresponding clause excludes the whole certified subcube and preserves every full root. Auxiliary literals must be retained when the witness depends on them; a short clause does not itself prove that a projected assignment has no witness.

One deletion pass, using a fresh restriction of the original generator for each test, yields an inclusion-minimal scalar-constant restriction in at most|S| tests. A failed deletion cannot become successful after further deletions, because restricting a constant more finely preserves it. This is not minimum cardinality: at the zero assignment, f=1+sum_e product_(i in e)x_i becomes constant exactly when the fixed-variable set hits every support e. The draft therefore identifies both a practical bounded search family and its optimization limit. No new general implicant-shrinking principle or global degree-of-regularity bound is claimed.

Validation custody is a separate prerequisite. The previous preparation stopped at the unchanged4MiB diagnostic allocation before any metadata phase, leaving manifest and builder/audit receipts absent. The proposed representation retains one ref map, explicit validated deltas, immutable commit/claims-tree results and actual command-output custody. It must reconstruct both every semantic map and any exact raw output it claims to preserve. Repeated claims-tree content can be represented once without changing the native highest-epoch/conflict decision, conditional on successful object reads, identical local records, stable parser/configuration and the same relevant time predicates. This is not equivalence of runtime successes, command failures or deadlines.

The byte formula counts initial state, deltas, unique tree entries, unique raw blobs, headers and reserved failure custody. Arbitrary churn can require full-size deltas; the design must refuse before crossing the existing cap. It makes no promise that all future repositories fit. This source/design step introduces no runtime helper or retry.

Next mathematical work is to identify cheaply certifiable projected-variable-only restriction sets in the actual quartic, membership-augmented and quadratic-lift encodings, while preserving complete projected sets, chart coverage, witness reconstruction and certificate checking costs. The reviewed compact-custody specification can support a separately admitted implementation if its proof and byte accounting survive critique. Neither path consumes the reserved independent scientific slot or resets the original two exhausted producer slots.

A symbolic order-dependence example makes the distinction concrete. For f=1+xy+xz at assignment(0,0,0), fixing x=0 gives constant1. Fixing y=z=0 also gives constant1, and neither y nor z can then be deleted. Deleting x first from the full assignment therefore yields{y,z}, whereas deleting y and then z first yields{x}. Both results are inclusion-minimal; only the latter has minimum cardinality for this example. Here D=2. With S={x}, the telescoping identity is f-1=x(y+z), and the conditional certificate is1=f-x(y+z), with product degree2. This is hand-derived algebra, not an executed fixture or benchmark.

The independent critic found no mathematical defect under the stated hypotheses and confirmed the native fence's content/epoch/conflict premise by source inspection. Three custody qualifications were incorporated: strict reversible ref-name handling, common boot/clock-domain timing with the native300-second freshness check, and conditional rather than full runtime equivalence. Concrete writer implementation, crash handling and byte-budget enforcement remain future obligations. The completed packet is a reviewed written design; no software correctness or measured performance acceptance is implied.

Sources: research/prime_scalar_compact_custody_design_20261108/mathematical-note.md (explicit identities, degree bounds and deletion proof); specification.md and proof-obligations.md in the same directory (custody premises and reconstruction); independent-static-review.md (exact input hashes and correction disposition). Native design decision DEC-20261108-17a496; source/design TASK-20261108-b69b89; snapshot TASK-20261108-d60cc4. Prior implicant work: Deharbe, Fontaine, Le Berre and Mazure, Computing prime implicants, FMCAD2013, https://www.verit-solver.org/papers/fmcad2013.pdf. This attribution locates prior shrinking/minimality work; no novelty claim is made here.